

Date	10 March 2026
Team Project name	Streamlining Ticket Assignment For Efficient Support Operations
Team ID	SWTID-2026-1605
Maximum mark	4 Marks

PERFORMANCE TESTING:

Performance testing is an important process used to evaluate how well the system performs under different conditions such as high user load, multiple ticket submissions, and simultaneous requests. The main purpose of performance testing is to ensure that the system works efficiently, responds quickly, and remains stable when many users access it at the same time.

In the proposed ticket management system, performance testing is conducted to measure the system's **speed, responsiveness, scalability, and stability**. Since the system handles multiple support tickets and users such as customers, support agents, and administrators, it is important to verify that the system can manage these operations without delays or failures.

One of the key aspects of performance testing is **response time**. Response time refers to the time taken by the system to process a user request and provide a response. For example, when a customer submits a support ticket, the system should quickly store the ticket in the database and generate a confirmation message. Similarly, when the system assigns a ticket to a support agent, the process should occur within a short time to ensure efficient support operations.

Another important factor is **system load handling**. The system must be capable of handling multiple users accessing the application simultaneously. During performance testing, several ticket requests are generated at the same time to evaluate how the system performs under heavy load conditions. This helps identify any performance bottlenecks and ensures that the system can handle large volumes of requests without slowing down.

Performance testing also evaluates the **database performance**. Since the system relies on a database to store ticket details, user information, and

status updates, it is important to ensure that data retrieval and storage operations are performed quickly. Efficient database queries and proper indexing help improve the speed of data processing and reduce system delays.

Another aspect considered during testing is **scalability**. As the organization grows, the number of customers and support tickets may increase significantly. The system should be able to scale and maintain good performance even when the number of users increases. Performance testing helps determine whether the system can handle future growth without requiring major changes.

The testing process also focuses on **system stability**. Stability testing ensures that the application continues to function properly over a long period of time without crashing or experiencing failures. Continuous operations such as ticket creation, assignment, and status updates are simulated to verify the reliability of the system.

To conduct performance testing, several tools and techniques can be used to simulate multiple users and monitor system behavior. These tools help measure system performance metrics such as response time, throughput, and resource usage.

The results of performance testing help developers identify areas where improvements may be required. For example, optimizing database queries, improving server configuration, or refining application logic can enhance the overall performance of the system.

In conclusion, performance testing ensures that the ticket management system operates efficiently under different workloads and user conditions. By analyzing system speed, scalability, and stability, performance testing helps ensure that the system provides reliable and fast support services to users. This ultimately leads to improved efficiency in support operations and better customer satisfaction.